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ORIGINAL PAPER

Maximal numerical ranges of certain classes of operators and approximation

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Abstract
Let $\mathcal{B}(\mathcal{H})$ be the collection of bounded linear operators on a complex separable Hilbert space $\mathcal{H}$. For $T \in \mathcal{B}(\mathcal{H})$, its numerical range and maximal numerical range are denoted by $W(T)$ and $W_0(T)$, respectively. First, we give in this paper a characterization of the maximal numerical range and, as applications, we determine maximal numerical ranges of weighted shifts, partial isometries, the Volterra integral operator and classical Toeplitz operators. Second, we study the universality of maximal numerical ranges, showing that any nonempty bounded convex closed subset of $\mathbb{C}$ is the maximal numerical range of some operator. Finally, we discuss the relations among the numerical range, the maximal numerical range and the spectrum. It is shown that the collection of those operators T with $W_0(T) \cap \sigma(T) = \emptyset$ is a nonempty open subset of $\mathcal{B}(\mathcal{H})$ precisely when $\dim \mathcal{H} > 1$, and is dense precisely when $1 < \dim \mathcal{H} < \infty$. We also show that those operators T with $W_0(T) = W(T)$ constitute a nowhere dense subset of $\mathcal{B}(\mathcal{H})$ precisely when $\dim \mathcal{H} > 1$

Keywords Maximal numerical range · Numerical range · Weighted shift · Partial isometry · Toeplitz operator

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1 Introduction

Throughout this paper, we denote by $\mathcal{H}$ a complex separable Hilbert space endowed with the inner product $\langle \cdot, \cdot \rangle$, and by $\mathcal{B}(\mathcal{H})$ the algebra of all bounded linear operators on $\mathcal{H}$. In this paper, an operator will always mean a bounded linear operator.

Recall that the *numerical range* of an operator A on $\mathcal{H}$ is defined by

$$W(A) := \{\langle Ax, x\rangle : x \in \mathcal{H}, \|x\| = 1\}.$$

The *numerical radius* of A is $w(A) := \sup\{|z| : z \in W(A)\}$. The notion of numerical range arose from the study of quadratic forms. The study of the numerical range and the numerical radius has a long history of over one hundred years. The classical Toeplitz-Hausdorff theorem (see [10, 18]) asserts that the numerical range of an operator is convex. Note that the numerical range of an operator is not necessarily closed except that its rank is finite.

In [17], Stampfli introduced the notion of maximal numerical range. The *maximal numerical range* of $A \in \mathcal{B}(\mathcal{H})$ is defined as

$$W_0(A) := \{\lambda \in \mathbb{C} : \langle Ax_n, x_n\rangle \to \lambda, \text{ where } \|x_n\| = 1 \text{ and } \|Ax_n\| \to \|A\|\}. \quad (1.1)$$

By [17], $W_0(A)$ is a nonempty convex, closed subset of the closure of $W(A)$. The notion of maximal numerical range plays a key role in determining norms of Rosenblum operators. For $A, B \in \mathcal{B}(\mathcal{H})$, the *Rosenblum operator* $\tau_{A,B}$ acting on $\mathcal{B}(\mathcal{H})$ is defined as $X \longmapsto AX - XB$. By [17], $0 \in W_0(A)$ is sufficient and necessary for an operator A to satisfy $\|\tau_{A,A}\| = 2\|A\|$; in addition, $0 \in W_0(A)$ if and only if $\|A\| \leq \|A + \lambda\|$ for all $\lambda \in \mathbb{C}$ (in this case, A is said to be *orthogonal* to the identity operator I in the Birkhoff-James sense [12]).

Recently there has been growing interest in maximal numerical ranges. In [9], Hamed and Spitkovsky explored the maximal numerical range for matrices and provided certain sufficient conditions for a matrix A to satisfy $W_0(A) = W(A)$. In particular, it was pointed out that $W_0(A)$ is exactly the numerical range of the compression of A onto the eigenspace of A^*A corresponding to its largest eigenvalue. In [16], Spitkovsky proved that $W_0(A) \cap \partial W(A) \neq \emptyset$ if and only if A is *normaloid*, that is, $\|A\|$ coincides with the spectral radius of A. In particular, when A is subnormal, it was shown that $W_0(A) = \text{conv}\{z \in \sigma(A) : |z| = \|A\|\}$, where conv denotes the convex hull. Later, Baghdad and Chraibi Kaadoud [2] extended the preceding result to hyponormal operators. The reader is referred to [1, 3, 13, 19] for more results on maximal numerical ranges.

The purpose of this paper is to provide a characterization of maximal numerical ranges and, as applications, answer some basic questions concerning maximal numerical ranges. To state our main results, we first introduce some notation and terminology.

Let $T \in \mathcal{B}(\mathcal{H})$. The unique positive square root of T^*T is denoted as $|T|$. We denote by $\ker T$ and $\text{ran } T$ the kernel of T and the range of T respectively. If $\mathcal{N}$ is a closed subspace of $\mathcal{H}$ and is invariant under T, then we denote by $T|_{\mathcal{N}}$ the restriction of T to $\mathcal{N}$, that is, $T|_{\mathcal{N}} \in \mathcal{B}(\mathcal{N})$ and $T|_{\mathcal{N}}(x) = Tx$ for $x \in \mathcal{N}$. Let $\mathcal{M}$ be a closed subspace of $\mathcal{H}$ and P be the orthogonal projection acting on $\mathcal{H}$ with $\text{ran } P = \mathcal{M}$. We

denote by $T_{\mathcal{M}}$ the *compression* of T to $\mathcal{M}$, that is, $T_{\mathcal{M}} = (PT)|_{\mathcal{M}}$. As usual, the closure of a subset Γ of $\mathbb{C}$ is denoted by Γ^-, card Γ denotes the cardinality of Γ and conv Γ stands for the convex hull of Γ.

The first result of this paper is the following theorem, which provides a useful description of the maximal numerical range of an operator in terms of numerical ranges.

Theorem 1.1 *Let $A \in \mathcal{B}(\mathcal{H})$ and E be the projection-valued spectral measure for $|A|$. Then $W_0(A) = \cap_{n\geq 1} W(A_{\mathcal{H}_n})^-$, where $\mathcal{H}_n = \text{ran } E([\|A\| - \frac{1}{n}, \|A\|])$.*

Remark 1.2 Just like numerical ranges, the convexity of the maximal numerical range is not obvious from its definition (1.1), and its proof is provided by Stampfli in [17, Lemma 2]. By Theorem 1.1, the convexity of maximal numerical ranges follows readily from that of numerical ranges.

As an application of Theorem 1.1, we shall determine maximal numerical ranges of weighted shifts in Sect. 2. A *weighted shift* operator T on a separable infinite dimensional Hilbert space $\mathcal{H}$ is an operator that maps each vector in some orthonormal basis $\{e_n\}$ into a scalar multiple of the next vector, i.e., $Te_n = w_n e_{n+1}$ for all n. If the index n runs over the set $\mathbb{N}$ of all nonnegative integers, then T is called a *unilateral weighted shift*; while if n runs over the set $\mathbb{Z}$ of all integers, then T is called a *bilateral weighted shift*.

To state our result, we need an extra notation. Given a nonempty subset Λ of $\mathbb{Z}$, we denote

$$\Phi(\Lambda) = \sup\{n \in \mathbb{N} \setminus \{0\} : \exists i \in \mathbb{Z} \text{ s.t. } i+1, i+2, \cdots, i+n \in \Lambda\}.$$

Clearly, $\Lambda_1 \subseteq \Lambda_2$ implies $\Phi(\Lambda_1) \leq \Phi(\Lambda_2)$.

For $\eta = (\eta_i)_{i\in\mathbb{Z}} \in l^\infty(\mathbb{Z})$, denote

$$\Delta(\eta) = \lim_n \Phi\Big(\{i \in \mathbb{Z} : \|\eta\|_\infty - \frac{1}{n} \leq |\eta_i| \leq \|\eta\|_\infty\}\Big).$$

The limit above exists because, for any $n \geq 1$,

$$\{i \in \mathbb{Z} : \|\eta\|_\infty - \frac{1}{n+1} \leq |\eta_i| \leq \|\eta\|_\infty\} \subset \{i \in \mathbb{Z} : \|\eta\|_\infty - \frac{1}{n} \leq |\eta_i| \leq \|\eta\|_\infty\}.$$

Similarly, one can define $\Delta(\eta)$ for $\eta = (\eta_i)_{i=1}^\infty \in l^\infty(\mathbb{N})$.

The following result determines maximal numerical ranges of weighted shifts.

Theorem 1.3 *Let $T \in \mathcal{B}(\mathcal{H})$ be a weighted shift with weights $\eta = (\eta_i)_{i\in\Lambda}$ and $\|T\| = 1$, where $\Lambda = \mathbb{N}$ or $\Lambda = \mathbb{Z}$.*

(i) If $\Delta(\eta) = n < \infty$, then $W_0(T) = \{z \in \mathbb{C} : |z| \leq \cos\frac{\pi}{n+1}\}$.
(ii) If $\Delta(\eta) = \infty$, then $W_0(T) = \{z \in \mathbb{C} : |z| \leq 1\}$.

As applications of the preceding result, we shall determine maximal numerical ranges of certain special weighted shifts, including the Kakutani shift, Donoghue shifts and the Bergman shift (see Corollary 2.6).

In the study of numerical ranges, it is difficult to determine for which convex subsets Ω of $\mathbb{C}$ there exists an operator A with $W(A) = \Omega$. By [20, Corollary 1.2], there exists no linear operator T so that $W(T) = \{z \in \mathbb{C} : |z| \leq 1\} \cup \{e^{i\theta} : \theta \in [0, 2\pi] \text{ is irrational}\}$. Bourin [4, Propositin 1.3] proved that those operators with closed numerical ranges are norm dense in $\mathcal{B}(\mathcal{H})$.

In Sect. 2, we shall describe maximal numerical ranges of partial isometries and, as an application, prove the following result, which shows that the maximal numerical range of an operator can be any nonempty bounded convex closed subset of $\mathbb{C}$.

Theorem 1.4 *Given a nonempty bounded convex closed subset Ω of $\mathbb{C}$, there exists a bounded linear operator A such that $W_0(A) = \Omega$.*

For any operator A, it is well known that $\sigma(A) \subseteq W(A)^-$. However, it is possible that $\sigma(A) \cap W(A) = \emptyset$ (see [8, Problem 215]). It is natural to discuss the relation between $\sigma(A)$ and $W_0(A)$ for $A \in \mathcal{B}(\mathcal{H})$. In fact, it is also possible that $\sigma(A) \cap W_0(A) = \emptyset$ (see Examples 2.4, 3.1 and 3.2).

Concerning the relation between maximal numerical ranges and spectra, we shall prove in Sect. 3 the following result.

Theorem 1.5 *Let $\mathcal{E} = \{A \in \mathcal{B}(\mathcal{H}) : \sigma(A) \cap W_0(A) = \emptyset\}$. Then*

(i) $\mathcal{E}$ is a nonempty open subset of $\mathcal{B}(\mathcal{H})$ precisely when $\dim \mathcal{H} > 1$;
(ii) $\overline{\mathcal{E}} = \mathcal{B}(\mathcal{H})$ if and only if $1 < \dim \mathcal{H} < \infty$.

By (1.1), if $A \in \mathcal{B}(\mathcal{H})$, then $W_0(A) \subseteq W(A)^-$. However, it is possible that $W_0(A) \cap W(A) = \emptyset$. For example, if A is the operator on $L^2[0, 1]$ given by $f(t) \mapsto tf(t)$, then $W_0(A) = \{1\}$ and $W(A) = (0, 1)$. It is natural to discuss the relation between $W_0(A)$ and $W(A)$ for general $A \in \mathcal{B}(\mathcal{H})$.

In Sect. 3, we shall prove the following result, which shows that there often exists a big gap between $W(A)$ and $W_0(A)$ for $A \in \mathcal{B}(\mathcal{H})$.

Theorem 1.6 *If $\dim \mathcal{H} > 1$, then the complement of $\{A \in \mathcal{B}(\mathcal{H}) : \text{card } W_0(A) = 1 < \text{card } W(A)\}$ is nowhere dense in $\mathcal{B}(\mathcal{H})$.*

The rest of this paper is organized as follows. The proofs of Theorems 1.1, 1.3 and 1.4 will be provided in Sect. 2. In the concluding section, we shall prove Theorems 1.5 and 1.6.

2 Proofs of Theorems 1.1, 1.3 and 1.4

This section is devoted to the proofs of Theorems 1.1, 1.3 and 1.4.

2.1 Proof of Theorem 1.1

Proof of Theorem 1.1 Clearly we may assume that $A \neq 0$. We denote by B_n and $\hat{B}_n$ the restrictions of $|A|$ to $\mathcal{H}_n$ and $\mathcal{H}_n^{\perp}$, respectively. Clearly, B_n and $\hat{B}_n$ are both positive. The proof is reduced to proving two claims.

Claim 1. $\cap_{n\geq 1} W(A_{\mathcal{H}_n})^- \subseteq W_0(A)$.

Arbitrarily choose $\lambda \in \cap_{n\geq 1} W(A_{\mathcal{H}_n})^-$. Then we can find unit vectors $\{\xi_k : k = 1, 2, 3, \cdots\} \subset \mathcal{H}$ with $\xi_k \in \mathcal{H}_k$ such that $|\langle A\xi_k, \xi_k\rangle - \lambda| < 1/k$ for all $k = 1, 2, 3, \cdots$. Note that

$$\sigma(B_k) = [\|A\| - \frac{1}{k}, \|A\|] \cap \sigma(|A|), \quad \sigma(B_k^{-1}) \subseteq [\frac{1}{\|A\|}, \frac{1}{\|A\| - \frac{1}{k}}].$$

Then

$$\|A\| \geq \|A\xi_k\| = \||A|\xi_k\| = \|B_k\xi_k\| \geq \|B_k^{-1}\|^{-1} \geq \|A\| - \frac{1}{k}, \quad k = 1, 2, 3, \cdots.$$

It follows that $\|A\xi_k\| \to \|A\|$ as $k \to \infty$. Recall that $\langle A\xi_k, \xi_k\rangle \to \lambda$. Therefore we obtain $\lambda \in W_0(A)$.

Claim 2. $W_0(A) \subseteq \cap_{n\geq 1} W(A_{\mathcal{H}_n})^-$.

Assume that $\lambda \in W_0(A)$. Now fix a positive integer k with $2k\|A\| > 1$. We shall show $\lambda \in W(A_{\mathcal{H}_k})^-$.

Since $\lambda \in W_0(A)$, by the definition, there exist unit vectors $\{\xi_n : n = 1, 2, 3, \cdots\} \subseteq \mathcal{H}$ such that $\|A\xi_n\| \to \|A\|$ and $\langle A\xi_n, \xi_n\rangle \to \lambda$ as $n \to \infty$. For $n \in \mathbb{N}$, assume $\xi_n = x_n + y_n$, where $x_n \in \mathcal{H}_k$ and $y_n \in \mathcal{H}_k^{\perp}$.

We claim that $y_n \to 0$ as $n \to \infty$. Indeed, since $\sigma(\hat{B}_k) \subseteq [0, \|A\| - \frac{1}{k}]$, it follows that

$$\begin{aligned}
\|A\xi_n\|^2 = \||A|\xi_n\|^2 &= \|B_k x_n\|^2 + \|\hat{B}_k y_n\|^2 \\
&\leq \|B_k\|^2\|x_n\|^2 + \|\hat{B}_k\|^2\|y_n\|^2 \\
&\leq \|A\|^2\|x_n\|^2 + (\|A\| - \frac{1}{k})^2\|y_n\|^2 \\
&= \|A\|^2 - \frac{1}{k}(2\|A\| - \frac{1}{k})\|y_n\|^2 \leq \|A\|^2.
\end{aligned}$$

Noting that $\|A\xi_n\| \to \|A\|$, this implies $y_n \to 0$ as $n \to \infty$. So $\|x_n\| \to 1$ and $\|Ax_n\| \to \|A\|$.

For each n large enough, denote $\eta_n = \frac{x_n}{\|x_n\|}$. Then $\eta_n \in \mathcal{H}_k$ is a unit vector and, as $n \to \infty$, we have

$$\|A\eta_n\| = \|Ax_n\|/\|x_n\| \to \|A\|$$

and

$$|\langle A\eta_n, \eta_n\rangle - \lambda| \leq |\langle A\eta_n, \eta_n\rangle - \langle A\xi_n, \xi_n\rangle| + |\langle A\xi_n, \xi_n\rangle - \lambda| \to 0.$$

It follows that $\lambda \in W(A_{\mathcal{H}_k})^-$. Thus we have proved that $\lambda \in W(A_{\mathcal{H}_n})^-$ for all integer n with $n > \frac{1}{2\|A\|}$. Since $\{W(A_{\mathcal{H}_n})^-\}_{n=1}^{\infty}$ is decreasing, one can see that Claim 2 holds. □

Remark 2.1 Let $A \in \mathcal{B}(\mathcal{H})$, E be the projection-valued spectral measure for $|A|$ and $\mathcal{H}_n = \text{ran } E([\|A\| - \frac{1}{n}, \|A\|])$ for $n \geq 1$. Obviously, $\|A\| \in \sigma(|A|)$. If $\|A\|$ is an isolated point of $\sigma(|A|)$, then it is clear that $\mathcal{H}_n = \ker(|A| - \|A\|)$ for n large enough and $W_0(A) = W(A_{\ker(|A|-\|A\|)})^-$.

By Theorem 1.1, the following result is clear.

Corollary 2.2 *Let $N \in \mathcal{B}(\mathcal{H})$ be a normal operator. Then,*

$$W_0(N) = \cap_{n\geq 1} W(N_{\mathcal{H}_n})^- = \text{conv}\,\{z \in \sigma(N) : |z| = \|N\|\},$$

where $\mathcal{H}_n = \text{ran } E(\{z \in \sigma(N) : \|N\| - \frac{1}{n} \leq |z| \leq \|N\|\})$ and E is the projection-valued spectral measure for N.

The preceding result was first proved by Spitkovsky [16, Theorem 2] and extended to subnormal operators. Baghdad and Chraibi Kaadoud [2, Theorem 3.3] further proved that $W_0(T) = \text{conv}\,\{z \in \sigma(T) : |z| = \|T\|\}$ when T is hyponormal. Next we shall show this also holds for Toeplitz operators on the classical Hardy space $H^2(\partial\mathbb{D})$, where $\mathbb{D}$ is the open unit disk in $\mathbb{C}$.

Proposition 2.3 *Let P be the projection of $L^2(\partial\mathbb{D})$ onto $H^2(\partial\mathbb{D})$. For $\phi \in L^\infty(\partial\mathbb{D})$, the Toeplitz operator T_ϕ on $H^2(\partial\mathbb{D})$ is defined by $T_\phi f = P(\phi f)$ for $f \in H^2(\partial\mathbb{D})$. Then the following three sets coincide*

$$W_0(T_\phi), \quad \text{conv}\,\{z \in \sigma(T_\phi) : |z| = \|T_\phi\|\}, \quad \text{conv}\,\{z \in \text{essran}\,\phi : |z| = \|T_\phi\|\}, \quad (2.1)$$

where essran ϕ *stands for the essential range of ϕ.*

Proof Let Ω_1, Ω_2 and Ω_3 denote those sets in (2.1) respectively. Let M_ϕ be defined by $M_\phi f = \phi f$ for $f \in L^2(\partial\mathbb{D})$. Since M_ϕ is normal, by [6, Corollary 7.8], we have $\|M_\phi\| = \|\phi\|_\infty = \|T_\phi\|$. It follows from Corollary 2.2 that

$$W_0(M_\phi) = \text{conv}\,\{z \in \text{essran}\,\phi : |z| = \|T_\phi\|\} = \Omega_3.$$

"$\Omega_1 \subset \Omega_3$". Choose $\lambda \in W_0(T_\phi)$. Then there exists $\{f_n\}_{n\geq 1} \subseteq H^2(\partial\mathbb{D})$ with $\|f_n\| = 1$ such that

$$\|M_\phi f_n\| \geq \|T_\phi f_n\| \to \|T_\phi\|, \quad \langle M_\phi f_n, f_n\rangle = \langle T_\phi f_n, f_n\rangle \to \lambda$$

as $n \to \infty$. So $\lambda \in W_0(M_\phi)$, that is, $W_0(T_\phi) \subseteq W_0(M_\phi) = \Omega_3$.

"$\Omega_3 \subset \Omega_2 \subset \Omega_1$". Recall that an operator R is *normaloid* if and only if its norm coincides with its spectral radius. Note that T_ϕ is normaloid and essran $\phi \subseteq \sigma(T_\phi)$ (see [6, Corollaries 7.7 & 7.8]). By [16, Theorem 1], we have

$$\text{conv}\,\{z \in \text{essran}\,\phi : |z| = \|T_\phi\|\} \subseteq \text{conv}\,\{z \in \sigma(T_\phi) : |z| = \|T_\phi\|\} \subseteq W_0(T_\phi).$$

That is, $\Omega_3 \subset \Omega_2 \subset \Omega_1$. □

As an application of Theorem 1.1, we next calculate the maximal numerical range of the Volterra integral operator.

For $e, f \in \mathcal{H}$, we denote by $e \otimes f$ the rank-one operator given by $[e \otimes f](x) = \langle x, f\rangle e$ for $x \in \mathcal{H}$.

Example 2.4 The Volterra integral operator on $L^2[0, 1]$ is defined as

$$(Vf)(x) = \int_0^x f(y)\mathrm{d}y, \quad x \in [0, 1],$$

where $f \in L^2[0, 1]$. We shall show $W_0(V) = \{\frac{4}{\pi^2}\}$.

By [8, Problem 188], we have

$$V^*V = \sum_{k=0}^{\infty} \frac{1}{(k + 1/2)^2\pi^2} f_k \otimes f_k,$$

where $f_k = \frac{1}{\sqrt{2}}(e^{\mathrm{i}\pi(k+1/2)x} + e^{-\mathrm{i}\pi(k+1/2)x})$, $k = 0, 1, 2, \cdots$. So $\||V|\| = \frac{2}{\pi}$ and $\frac{2}{\pi}$ is an isolated point of $\sigma(|V|)$. Denote $\widetilde{\mathcal{H}} = \ker(\frac{2}{\pi} - |V|)$. By Remark 2.1, we have $W_0(V) = W(V_{\widetilde{\mathcal{H}}})^-$. Note that $\ker(\frac{2}{\pi} - |V|) = \mathbb{C}f_0$ and

$$\begin{aligned}\langle Vf_0, f_0\rangle &= \int_0^1 \int_0^x f_0(y)\overline{f_0(x)}\mathrm{d}y\mathrm{d}x \\ &= \int_0^1 \int_0^x (\sqrt{2}\cos\frac{\pi}{2}y)(\sqrt{2}\cos\frac{\pi}{2}x)\mathrm{d}y\mathrm{d}x = \frac{4}{\pi^2}.\end{aligned}$$

Hence we obtain $W_0(V) = \{\frac{4}{\pi^2}\}$.

2.2 Proof of Theorem 1.3

To give the proof of Theorem 1.3, we need an extra lemma.

Lemma 2.5 *Let A and B be (unilateral or bilateral) weighted shifts with nonnegative weights $\{a_n\}$ and $\{b_n\}$ (relatively to the same orthonormal basis), respectively. If $a_n \le b_n$ for all n, then $W(A)^- \subseteq W(B)^-$.*

Proof We only prove the result for the unilateral case. The proof for the bilateral case is similar. Assume that $\{f_i\}_{i\in\mathbb{N}}$ is an orthonormal basis of $\mathcal{H}$, $Af_i = a_i f_{i+1}$ and $Bf_i = b_i f_{i+1}$ for $i \in \mathbb{N}$. By [15, page 72], it suffices to prove $w(A) \le w(B)$.

For any $x = \sum_{i\in\mathbb{N}} \alpha_i f_i \in \mathcal{H}$ with $\|x\| = 1$, we denote $\widetilde{x} = \sum_{i\in\mathbb{N}} |\alpha_i| f_i$. Then

$$|\langle Ax, x\rangle| \le \langle A\widetilde{x}, \widetilde{x}\rangle \le \langle B\widetilde{x}, \widetilde{x}\rangle \le w(B).$$

This implies $w(A) \le w(B)$. □

It is easy to see that the preceding result also holds for truncated weighted shifts.

For n a positive integer, we denote by J_n the operator on $\mathbb{C}^n$ which admits the following matrix representation

$$J_n = \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{bmatrix}_{n\times n}$$

with respect to the canonical orthonormal basis of $\mathbb{C}^n$.

Now we can give the proof of Theorem 1.3.

Proof of Theorem 1.3 We just give the proof in the case that $\Lambda = \mathbb{Z}$. The proof for the other case is similar.

We assume that $\{f_i\}_{i\in\mathbb{Z}}$ is an orthonormal basis of $\mathcal{H}$ and $Tf_i = \eta_i f_{i+1}$ for $i \in \mathbb{Z}$. Also, by [15, Page 52, Corollary 1], we may assume that $\eta_i \ge 0$ for all $i \in \mathbb{Z}$. It is clear that

$$|T| = \sum_{i\in\mathbb{Z}} \eta_i f_i \otimes f_i.$$

Given a nonempty subset Γ of $\mathbb{Z}$, we denote $\mathcal{L}_\Gamma := \vee\{f_i : i \in \Gamma\}$, where $\vee$ denotes the closed linear span. For each $n \ge 1$, denote

$$\Lambda_n = \{i \in \mathbb{Z} : 1 - \frac{1}{n} \le \eta_i \le 1\}.$$

Recall that $\Delta(\eta) = \lim_n \Phi(\Lambda_n)$. It is easy to see

$$\mathcal{H}_n = \text{ran } E([\|T\| - \frac{1}{n}, \|T\|]) = \vee\{f_i : i \in \Lambda_n\},$$

where E is the projection-valued spectral measure of $|T|$.

Case 1. $\Delta(\eta) = m < \infty$.

This implies that there exists $N > 0$ such that $\Phi(\Lambda_n) = m$ for all $n \ge N$.

Now we fix n with $n \ge N$. Then there is a partition $\{\Gamma_k\}_{k\in\Upsilon}$ of Λ_n satisfying the following:

(a) For each $k \in \Upsilon$, $i, j \in \Gamma_k$ implies $s \in \Gamma_k$ for any $i \le s \le j$.
(b) $\text{dist}(\Gamma_{k_1}, \Gamma_{k_2}) := \inf\{|i - j| : i \in \Gamma_{k_1}, j \in \Gamma_{k_2}\} \ge 2$ whenever $k_1 \ne k_2$.

For each $k \in \Upsilon$, denote

$$n_k = \operatorname{card} \Gamma_k, \quad \mathcal{H}_n = \bigoplus_{k\in\Upsilon} \mathcal{L}_{\Gamma_k}, \quad S_k = T_{\mathcal{L}_{\Gamma_k}}.$$

Then $T_{\mathcal{H}_n} = \oplus_{k\in\Upsilon} S_k$.

Since $\Phi(\Lambda_n) = m$, it follows that $\max_{k\in\Upsilon} n_k = m$. Without loss of generality, we assume $n_1 = m$. By Lemma 2.5,

$$\begin{aligned}(1-\frac{1}{n})W(J_m) \subseteq W(S_1)^- &\subseteq W(T_{\mathcal{H}_n})^- \\ &\subseteq \operatorname{conv}[\cup_{k\in\Upsilon} W(S_k)]^- \\ &\subseteq W(J_m).\end{aligned}$$

That is, $(1-\frac{1}{n})W(J_m) \subseteq W(T_{\mathcal{H}_n})^- \subseteq W(J_m)$.

Note that $\{W(T_{\mathcal{H}_n})^-\}_{n=1}^{\infty}$ is decreasing and $\{(1-\frac{1}{n})W(J_m)\}_{n=1}^{\infty}$ is increasing. It follows immediately that

$$\bigcup_{n\geq N}(1-\frac{1}{n})W(J_m) \subseteq W_0(T) = \cap_{n=N}^{\infty} W(T_{\mathcal{H}_n})^- \subseteq W(J_m).$$

Hence $W_0(T) = W(J_m)$. By [7, Proposition 1], $W(J_m) = \{z \in \mathbb{C} : |z| \leq \cos\frac{\pi}{m+1}\}$. Then the desired result follows readily.

Case 2. $\Delta(\eta) = \infty$.

Since $\|T\| = 1$, it is clear that $W_0(T) \subseteq \{z \in \mathbb{C} : |z| \leq 1\}$. So we only need to prove the converse inclusion.

Since $\Delta(\eta) = \infty$, it follows that there exists $N > 0$ such that $\Phi(\Lambda_n) = \infty$ for all $n \geq N$. Then, for any $n \geq N$ and any $m \in \mathbb{N}$, there exists $r \in \mathbb{Z}$ such that

$$\{\eta_r, \eta_{r+1}, \cdots, \eta_{r+m-1}\} \subset [1-\frac{1}{n}, 1].$$

Denote $\mathcal{L}_{n,m} = \vee\{f_{r+i} : 0 \leq i \leq m-1\}$. Then $\mathcal{L}_{n,m} \subseteq \mathcal{H}_n$. Define

$$R_{n,m} = \begin{bmatrix} 0 & & & & \\ 1-\frac{1}{n} & 0 & & & \\ & 1-\frac{1}{n} & 0 & & \\ & & \ddots & \ddots & \\ & & & 1-\frac{1}{n} & 0 \end{bmatrix} \begin{matrix} f_r \\ f_{r+1} \\ f_{r+2} \\ \vdots \\ f_{r+m-1} \end{matrix}.$$

Then, by Lemma 2.5, we obtain

$$(1-\frac{1}{n})W(J_m) = W(R_{n,m})^- \subseteq W(T_{\mathcal{L}_{n,m}})^- \subseteq W(T_{\mathcal{H}_n})^-.$$

Since m is arbitrary,

$$\{z \in \mathbb{C} : |z| < 1 - \frac{1}{n}\} = \bigcup_{m \geq 1} [(1 - \frac{1}{n}) W(J_m)] \subseteq W(T_{\mathcal{H}_n})^-.$$

By Theorem 1.1, we have

$$\{z \in \mathbb{C} : |z| < 1 - \frac{1}{n}\} \subseteq \bigcap_{l \geq n} W(T_{\mathcal{H}_l})^- = W_0(T).$$

Since n was arbitrary and $W_0(T)$ is closed, we deduce that $\{z \in \mathbb{C} : |z| \leq 1\} \subset W_0(T)$ and, furthermore, $\{z \in \mathbb{C} : |z| \leq 1\} = W_0(T)$. □

Corollary 2.6 *(i) If T is the Kakutani shift, that is, T is the unilateral weighted shift with weights $\{\frac{1}{\gcd\ (i+1, 2^{i+1})}\}_{i=0}^{\infty}$, where* gcd *stands for the greatest common divisor, then $W_0(T) = \{0\}$.*

(ii) If T is a Donoghue shift with $\|T\| = 1$, that is, T is a unilateral weighted shift with decreasing, positive weights $\{\eta_i\}_{i=0}^{\infty}$ satisfying $\{\eta_i\}_{i=0}^{\infty} \in l^2$, then $W_0(T) = W(J_n) = \{z \in \mathbb{C} : |z| \leq \cos \frac{\pi}{n+1}\}$, where $n = \text{card}\ \{i \geq 0 : \eta_i = 1\}$.

(iii) If T is the Bergman shift, that is, T is the unilateral weighted shift with weights $\{\sqrt{\frac{n+1}{n+2}}\}_{n=0}^{\infty}$, then $W_0(T) = \{z \in \mathbb{C} : |z| \leq 1\}$.

Example 2.7 We let T denote the unilateral shift. For $\varepsilon > 0$, we denote by T_ε the unilateral weighted shift with weights $(1 + \varepsilon, 1, 1, 1, \cdots)$. Then, by Theorem 1.3, $W_0(T) = \{z \in \mathbb{C} : |z| \leq 1\}$ and $W_0(T_\varepsilon) = \{0\}$. This shows that the map $X \mapsto W_0(X)$ is not continuous.

2.3 Proof of Theorem 1.4

Proof of Theorem 1.4 Without loss of generality, we assume that $\Omega \subseteq \mathbb{D}$, where $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.

Choose a sequence $\{c_n\}_{n \geq 1} \subset \Omega$ such that $\{c_n : n \geq 1\}^- = \Omega$. Define a diagonal operator $D = \text{diag}\{c_1, c_2, c_3, \cdots\}$ acting on $\mathcal{H}$. Then $W(D)^- = \Omega$. Set

$$U = \begin{bmatrix} D & 0 \\ \sqrt{I - D^*D} & 0 \end{bmatrix} \begin{matrix} \mathcal{H}_1 \\ \mathcal{H}_2 \end{matrix},$$

where $\mathcal{H}_1 = \mathcal{H}_2 = \mathcal{H}$ and I is the identity operator on $\mathcal{H}$. Clearly, U is a partial isometry. It suffices to show that $W_0(U) = W(D)^-$.

Since U^*U is an orthogonal projection, it is clear that

$$|U| = |U|^2 = U^*U = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \begin{matrix} \mathcal{H}_1 \\ \mathcal{H}_2 \end{matrix}.$$

Note that $1 = \|U\|$ is an isolated eigenvalue of $|U|$ and $\ker(I - |U|) = (\ker U)^{\perp} = \mathcal{H}_1$. Then, by Theorem 1.1 and Remark 2.1, we have

$$W_0(U) = W(U_{(\ker U)^{\perp}})^{-} = W(D)^{-} = \Omega,$$

which completes the proof. □

The following corollary can also be derived directly (and easily) from Theorem 1.1.

Corollary 2.8 *Let U be a partial isometry and A be the compression of U to its initial space $(\ker U)^{\perp}$. Then $W_0(U) = W(A)^{-}$.*

Next we provide a mild improvement of Theorem 1.4.

Proposition 2.9 *Given a nonempty bounded convex closed subset Ω of $\mathbb{C}$, there exists a normaloid bounded linear operator A such that $W_0(A) = \Omega$.*

Proof Without loss of generality, we assume that $\Omega \subseteq \mathbb{D}$. Then there is $\mu_0 \in \mathbb{C} \setminus \{0\}$ such that

$$\hat{\Omega} := \{\mu_0 \mu : \mu \in \Omega\} \subseteq \mathbb{D}^{-}, \ \hat{\Omega} \cap \partial\mathbb{D} \neq \emptyset.$$

By the proof of Theorem 1.4, we can find a partial isometry V such that $W_0(V) = \hat{\Omega}$ and $V|_{(\ker V)^{\perp}}$ is diagonal operator with norm 1. By [14, Theorem 3.1], V is normaloid and so is V/μ_0. Therefore $A := V/\mu_0$ is normaloid with $W_0(A) = \Omega$. □

Remark 2.10 The result of Proposition 2.9 is sharp in the sense that A can not be required to be hyponormal (or normal). For example, if $\Omega = [0, 1]$, then, by Corollary 2.2 and [16, Theorem 2]), there exists no normal or hyponormal operator A such that $W_0(A) = [0, 1]$.

3 Proofs of Theorems 1.5 and 1.6

In this section, the proofs of Theorems 1.5 and 1.6 will be given. We first introduce some notation and terminology.

An operator $T \in \mathcal{B}(\mathcal{H})$ is called a *semi-Fredholm operator*, if ran T is closed and either nul T or nul T^* is finite, where nul $T = \dim \ker T$ and nul $T^* = \dim \ker T^*$. In this case, ind T = nul T − nul T^* is called the *index* of T. In particular, if $-\infty < \text{ind } T < +\infty$, then T is called a *Fredholm operator*. The *essential spectrum* $\sigma_e(T)$ and *semi-Fredholm domain* of index n are defined by

$$\sigma_e(T) = \{\lambda \in \mathbb{C} : T - \lambda \text{ is not Fredholm}\},$$

$$\rho_{s-F}^{(n)}(T) = \{\lambda \in \mathbb{C} : T - \lambda \text{ is semi-Fredholm with ind } (T - \lambda) = n\},$$

respectively, where $-\infty \leq n \leq +\infty$.

3.1 Proof of Theorem 1.5

We first make some preparations.

The following two examples show that the maximal numerical range of an operator may not intersect its spectrum.

Example 3.1 Let $n \in \mathbb{N}$ with $n \geq 2$ and $T \in \mathcal{B}(\mathbb{C}^n)$ with $T = UP$, where

$$U = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & & & \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & & & \\ & & \mathrm{i} & & \\ & & & \ddots & \\ & & & & \mathrm{i} \end{bmatrix} \begin{matrix} e_1 \\ e_2 \\ e_3, \\ \vdots \\ e_n \end{matrix} \quad P = \begin{bmatrix} 2 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix} \begin{matrix} e_1 \\ e_2 \\ e_3, \\ \vdots \\ e_n \end{matrix}$$

and $\{e_i\}_{i=1}^n$ is the canonical orthonormal basis of $\mathbb{C}^n$. Clearly, U is unitary and $P = |T|$. Then T is invertible with $T = UP$ being its polar decomposition. We shall show $W_0(T) \cap \sigma(T) = \emptyset$.

Note that $\|T\| = 2$ and $\ker(|T| - 2) = \vee\{e_1\}$. Then, by Theorem 1.1, we have $W_0(T) = \{\langle Te_1, e_1\rangle\} = \{1\}$. On the other hand, note that

$$T - 1 = \begin{bmatrix} 0 & \frac{\sqrt{3}}{2} & & & \\ -\sqrt{3} & -\frac{1}{2} & & & \\ & & \mathrm{i} - 1 & & \\ & & & \ddots & \\ & & & & \mathrm{i} - 1 \end{bmatrix}$$

is invertible, that is, $1 \notin \sigma(T)$. Hence we conclude that $W_0(T) \cap \sigma(T) = \emptyset$.

Example 3.2 Let $A \in \mathcal{B}(\mathcal{H})$ be defined as

$$Ae_n = \begin{cases} e_{n+1}, & n \neq 0; \\ 2e_{n+1}, & n = 0. \end{cases}$$

where $\{e_n : n \in \mathbb{Z}\}$ is an orthonormal basis of $\mathcal{H}$. Then $\sigma(A) = \{z \in \mathbb{C} : |z| = 1\}$ and, by Theorem 1.3, it is easy to see $W_0(A) = \{0\}$.

Stampfli [17] proved that $W_0(A) \cap W_0(A + \lambda) = \emptyset$ for all $A \in \mathcal{B}(\mathcal{H})$ and all $\lambda \in \mathbb{C}\backslash\{0\}$. On the other hand, Example 2.7 shows that the map $T \mapsto W_0(T)$ is not lower semi-continuous. However, the following lemma shows that it is upper semi-continuous.

Lemma 3.3 *Given $A \in \mathcal{B}(\mathcal{H})$ and $\varepsilon > 0$, there exists $c > 0$ such that*

$$W_0(A + B) \subseteq W_0(A) + \mathrm{N}(0, \varepsilon)$$

for any $B \in \mathcal{B}(\mathcal{H})$ with $\|B\| < c$, where $\mathrm{N}(0, \varepsilon) = \{z \in \mathbb{C} : |z| \leq \varepsilon\}$.

Proof We first prove the following key claim.

Claim. For any closed half plane Ω such that $W_0(A) \subset \Omega^\circ$ (where Ω° denotes the interior of Ω), there exists $\delta > 0$ such that $W_0(A+B) \subset \Omega^\circ$ for all B with $\|B\| < \delta$.

Without loss of generality, we may assume that $\|A\| = 1$ and $\Omega = \{z \in \mathbb{C} : \text{Re } z \geq 0\}$. Set $\eta = \inf \text{Re } W_0(A)$. Then $\eta > 0$.

For each r with $0 < r < \eta$, define

$$E_r = \{x \in \mathcal{H} : \|x\| = 1, \text{Re } \langle Ax, x\rangle \leq \eta - r/4\}.$$

Then, by the definition of $W_0(A)$, $M_r := \sup_{x \in E_r} \|Ax\| < 1$ (if $E_\varepsilon = \emptyset$, we define $M_\varepsilon = 0$). Set $\delta = \min\{(1 - M_r^2)/8, r/4\}$. Next we shall show δ satisfies all requirements.

Fix an operator B with $\|B\| < \delta$. If $y \in E_r$, then $M_r^2 < 1 - 4\delta < 1 - 4\|B\|$ and

$$\begin{aligned}\|(A+B)y\| &\leq (\|Ay\|^2 + \|B\|^2 + 2\|B\|)^{\frac{1}{2}} \leq (M_r^2 + \|B\|^2 + 2\|B\|)^{\frac{1}{2}} \\ &< 1 - \|B\| \leq \|A+B\|.\end{aligned}$$

Hence, if $\mu \in W_0(A+B)$, then there exist unit vectors $\{y_n\}_{n=1}^\infty \subset \mathcal{H} \setminus E_r$ such that

$$\begin{aligned}\text{Re } \mu &= \lim_{n\to\infty} \text{Re } \langle (A+B)y_n, y_n\rangle \geq \liminf_{n\to\infty} \text{Re } \langle Ay_n, y_n\rangle - \|B\| \\ &\geq \eta - r/4 - r/2 > r/4 > 0.\end{aligned}$$

This implies that $W_0(A+B) \subset \Omega^\circ$, which proves the claim.

For any $\varepsilon > 0$, there exist closed half planes $\{\Omega_k\}_{k=1}^n$ such that

$$W_0(A) \subseteq \cap_{k=1}^n \Omega_k^\circ \subseteq W_0(A) + \text{N}(0, \varepsilon).$$

By Claim, we can find $c > 0$ such that

$$W_0(A+B) \subseteq \cap_{k=1}^n \Omega_k^\circ \subseteq W_0(A) + \text{N}(0, \varepsilon)$$

for any operator B with norm less than c. This completes the proof. □

The following example shows that the set $\{X \in \mathcal{B}(\mathcal{H}) : \sigma(X) \cap W_0(X) \neq \emptyset\}$ has a nonempty interior provided $\dim \mathcal{H} = \infty$.

Example 3.4 Let

$$A = S \oplus \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \in \mathcal{B}(\mathcal{H}),$$

where S is the unilateral shift. Next we shall show there exists $\delta > 0$ such that

$$W_0(A+K) \cap \sigma(A+K) \neq \emptyset$$

for all $K \in \mathcal{B}(\mathcal{H})$ with $\|K\| < \delta$.

Note that ind $(A - z) = -1$ for all $z \in \mathrm{N}(0, 1)$. Since $\rho_{s-F}^{(-1)}(\cdot)$ is lower semi-continuous (see [11, Theorem 1.14 (ii)]), there exits $\delta_0 > 0$ such that

$$\mathrm{N}(0, 1) = \rho_{s-F}^{(-1)}(A) \subseteq \rho_{s-F}^{(-1)}(A + K) + \mathrm{N}(0, \frac{1}{2}),$$

for any $K \in \mathcal{B}(\mathcal{H})$ with $\|K\| < \delta_0$. Then $\mathrm{N}(0, \frac{1}{2}) \subseteq \rho_{s-F}^{(-1)}(A + K) \subseteq \sigma(A + K)$.

On the other hand,

$$\|S\| = 1 < 2 = \left\| \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \right\| = \|A\|,$$

we have

$$W_0(A) = W_0\left(\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \right) = \{0\}.$$

Then, by Lemma 3.3, there exists $\delta < \delta_0$ such that $W_0(A + K) \subseteq \mathrm{N}(0, \frac{1}{2})$ for any $K \in \mathcal{B}(\mathcal{H})$ with $\|K\| < \delta$. Furthermore, $W_0(A + K) \cap \sigma(A + K) \neq \emptyset$.

The following result is elementary. For the reader's convenience, however, the proof is provided.

Lemma 3.5 *Let $\mathcal{U}(\mathbb{C}^n) := \{U \in \mathcal{B}(\mathbb{C}^n) : U \text{ is unitary}\}$ and $\{e_i\}_{i=1}^n$ be an orthonormal basis of $\mathbb{C}^n$. Then*

$$\mathcal{V}(\mathbb{C}^n) := \{U \in \mathcal{U}(\mathbb{C}^n) : \langle Ue_i, e_j \rangle \neq 0, \forall\, 1 \leq i, j \leq n\}$$

is an open, norm dense subset of $\mathcal{U}(\mathbb{C}^n)$.

Proof We just give the proof in the case that $n = 3$. The proof for the general case is similar.

For $1 \leq i, j \leq 3$, denote $\mathcal{W}_{i,j} := \{U \in \mathcal{U}(\mathbb{C}^3) : \langle Ue_i, e_j \rangle = 0\}$. Clearly, each $\mathcal{W}_{i,j}$ is closed and $\cup_{i,j} \mathcal{W}_{i,j}$ is the complement of $\mathcal{V}(\mathbb{C}^3)$. It suffices to prove that each $\mathcal{W}_{i,j}$ is nowhere dense in $\mathcal{U}(\mathbb{C}^3)$. We need only prove it in the case that $i = j = 1$.

Assume that $U \in \mathcal{W}_{1,1}$ with $U = [u_{i,j}]$. Then $u_{1,1} = 0$. Since U is unitary, we may assume that $u_{1,2} \neq 0$. Let

$$U_\theta = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} \\ u_{2,1} & u_{2,2} & u_{2,3} \\ u_{3,1} & u_{3,2} & u_{3,3} \end{bmatrix} \begin{bmatrix} \cos\,\theta & \sin\,\theta & 0 \\ -\sin\,\theta & \cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where $\theta \in (0, \frac{\pi}{2})$. Then $U_\theta \in \mathcal{U}(\mathbb{C}^3)$. Denote $U_\theta = [u_{i,j}(\theta)]$. Clearly,

$$\|U_\theta - U\| = \left\| \begin{bmatrix} \cos\,\theta - 1 & \sin\,\theta \\ -\sin\,\theta & \cos\,\theta - 1 \end{bmatrix} \right\| \to 0 \text{ as } \theta \to 0.$$

Then there exists sufficiently small $\theta \in (0, \frac{\pi}{2})$ such that $u_{1,1}(\theta) \neq 0$. Thus $\mathcal{W}_{1,1}$ is nowhere dense, which completes the proof. □

Similarly one can see the following result.

Corollary 3.6 *Let $\mathcal{U}(\mathcal{H})$ be the set consisting of all unitary operators on $\mathcal{H}$ and $\{e_n\}_{n=1}^{\infty}$ be an orthonormal basis of $\mathcal{H}$. Then*

$$\mathcal{V}(\mathcal{H}) := \{U \in \mathcal{U}(\mathcal{H}) : \langle Ue_i, e_j\rangle \neq 0, \forall\, i, j \geq 1\}$$

is a second category subset of $\mathcal{U}(\mathcal{H})$.

Now we are going to give the proof of Theorem 1.5.

Proof of Theorem 1.5 (i) By the upper semi-continuity of the spectrum as well as the maximal numerical range (see Lemma 3.3), one can see that the set $\mathcal{E}$ is an open subset of $\mathcal{B}(\mathcal{H})$.

If $\dim \mathcal{H} = 1$, then it is trivial to see that $\mathcal{E}$ is empty. If $1 < \dim \mathcal{H} < \infty$, then, by Example 3.1, $\mathcal{E}$ is nonempty. If $\dim \mathcal{H} = \infty$, then, by Example 3.2, $\mathcal{E}$ is nonempty.

(ii) If $\dim \mathcal{H} = \infty$, then, by Example 3.4, one can see $\mathcal{E}^- \neq \mathcal{B}(\mathcal{H})$. Next we directly assume that $\mathcal{H} = \mathbb{C}^n$ with $1 < n < \infty$. We shall prove $\mathcal{E}^- = \mathcal{B}(\mathbb{C}^n)$. For briefness, we just give the proof in the case that $n = 3$. The proof for the general case is similar.

Assume that $B \in \mathcal{B}(\mathbb{C}^3)$. We may directly assume that $\|B\| < 1$. Since invertible operators are norm dense in $\mathcal{B}(\mathbb{C}^3)$, we may also assume that B is invertible.

Suppose that $B = UP$ is the polar decomposition of B, where U is unitary and $P = |B|$. Assume that U and P admit the following matrix representations

$$U = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} \\ u_{2,1} & u_{2,2} & u_{2,3} \\ u_{3,1} & u_{3,2} & u_{3,3} \end{bmatrix} \begin{matrix} f_1 \\ f_2, \\ f_3 \end{matrix} \quad P = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} \begin{matrix} f_1 \\ f_2 \\ f_3 \end{matrix}$$

with respect to the canonical orthonormal basis $\{f_i : i = 1, 2, 3\}$ of $\mathbb{C}^3$, respectively. We may also assume that $\lambda_1 > \lambda_2 > \lambda_3 > 0$ (at most up to an arbitrarily small perturbation).

Claim. Given $\varepsilon > 0$, there exists a unitary operator $V \in \mathcal{B}(\mathbb{C}^3)$ with $\|U - V\| < \varepsilon$ such that

(i) $\langle Vf_i, f_j\rangle \neq 0$ for all $1 \leq i, j \leq 3$, and
(ii) $\sum_{i=2}^{3} \langle Vf_1, f_i\rangle\langle Vf_i, f_1\rangle \neq 0$.

By Lemma 3.5, there exists unitary operator $W \in \mathcal{B}(\mathbb{C}^3)$ with $\|U - W\| < \varepsilon/2$ such that $\langle Wf_i, f_j\rangle \neq 0$ for all $1 \leq i, j \leq 3$.

If $\sum_{i=2}^{3} \langle Wf_1, f_i\rangle\langle Wf_i, f_1\rangle \neq 0$, then we set $V = W$. Otherwise, we let

$$V = W \begin{bmatrix} 1 & & \\ & 1 & \\ & & e^{\mathrm{i}\alpha} \end{bmatrix}$$

with some sufficient small α satisfying $0 < |e^{\mathrm{i}\alpha} - 1| < \varepsilon/2$. Clearly, we can find suitable α such that V satisfies all requirements. This proves the claim.

Assume that $V = [v_{i,j}]_{3\times 3}$ relative to $\{f_1, f_2, f_3\}$. For each $r > 0$, define

$$P_r = \begin{bmatrix} \lambda_1 + r & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} \begin{matrix} f_1 \\ f_2 \\ f_3 \end{matrix}$$

and $C_r = VP_r$. Then $\|B - C_r\| \le \|UP - VP\| + \|VP - VP_r\| < \varepsilon + r$. Next we shall show that there exists sufficiently small $r > 0$ such that $W_0(C_r) \cap \sigma(C_r) = \emptyset$.

Arbitrarily fix an $r > 0$. Note that $C_r = VP_r$ is the polar decomposition of C_r. Since $\lambda_1 + r > \lambda_2 > \lambda_3$, it follows immediately from Theorem 1.1 that $W_0(C_r) = \{(\lambda_1 + r)v_{11}\}$.

Note that the determinant δ_r of the following matrix

$$\begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \begin{bmatrix} \lambda_1 + r & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} - (\lambda_1 + r)v_{11} I$$

is of the form

$$\delta_r = -(\lambda_1 + r)\Big[v_{11}\Big(\sum_{i=2}^{3} v_{1i}v_{i1}\Big)r + c\Big]$$

for some constant c. Clearly, we can find sufficiently small $r \in (0, \varepsilon)$ such that $\delta_r \neq 0$, which implies that $(\lambda_1 + r)v_{11} \notin \sigma(C_r)$ and $W_0(C_r) \cap \sigma(C_r) = \emptyset$. Note that $\|B - C_r\| < 2\varepsilon$. Since ε is arbitrary, we conclude the proof. □

3.2 Proof of Theorem 1.6

To give the proof of Theorem 1.6, we make some extra preparations.

Let $T \in \mathcal{B}(\mathcal{H})$. If σ is a clopen subset of $\sigma(T)$ (that is, both σ and $\sigma(T) \setminus \sigma$ are closed subsets of $\sigma(T)$), then there exists an analytic Cauchy domain Ω such that $\sigma \subseteq \Omega$ and $[\sigma(T)\backslash\sigma] \cap \Omega^- = \emptyset$. We let $E(\sigma; T)$ denote the Riesz idempotent of T corresponding to σ, that is,

$$E(\sigma; T) = \frac{1}{2\pi \mathrm{i}} \int_\Gamma (\lambda - T)^{-1}\, \mathrm{d}\lambda,$$

where $\Gamma = \partial\Omega$ is positively oriented with respect to Ω in the sense of complex variable theory. In this case, we denote $\mathcal{H}(\sigma; T) = \operatorname{ran} E(\sigma; T)$. If λ is an isolated point of $\sigma(T)$, then $\{\lambda\}$ is a clopen subset of $\sigma(T)$ and we simply write $\mathcal{H}(\lambda; T)$ instead of $\mathcal{H}(\{\lambda\}; T)$. If, in addition, $\dim \mathcal{H}(\lambda; T) < \infty$, then λ is called a *normal eigenvalue* of T. Each normal eigenvalue of T lies in the point spectrum $\sigma_p(T)$. The set of all normal eigenvalues of T will be denoted by $\sigma_0(T)$. The reader is referred to [11, page 2] for more about normal eigenvalues.

Lemma 3.7 *[[11, Corollary 1.6]] Let $A, B \in \mathcal{B}(\mathcal{H})$, σ be a nonempty clopen subset of $\sigma(A)$ and a Cauchy domain Ω be a neighborhood of σ such that $[\sigma(A)\backslash\sigma] \cap \Omega^- = \emptyset$. If $\|A - B\| < \min\{\|(\lambda - A)^{-1}\|^{-1} : \lambda \in \partial\Omega\}$, then*

(i) $\sigma' := \sigma(B) \cap \Omega \neq \emptyset$, $\sigma(B) \cap \partial\Omega = \emptyset$, *and*
(ii) $\dim \mathcal{H}(\sigma; A) = \dim \mathcal{H}\left(\sigma'; B\right)$.

Recall that two operators $A, B \in \mathcal{B}(\mathcal{H})$ are said to be *approximately unitarily equivalent*, denoted by $A \cong_a B$, if there exist unitary operators $\{U_n\}_{n=1}^{\infty}$ such that $AU_n - U_n B \to 0$ as $n \to \infty$, or equivalently, the closures of the unitary equivalence classes of A and B coincide.

Now we can give the proof of Theorem 1.6.

Proof of Theorem 1.6 First, denote

$$\mathcal{F} = \{A \in \mathcal{B}(\mathcal{H}) : \text{card } W_0(A) = 1 < \text{card } W(A)\}.$$

It is clear that

$$\mathcal{F}^c = \{A \in \mathcal{B}(\mathcal{H}) : \text{card } W_0(A) > 1\} \cup \{A \in \mathcal{B}(\mathcal{H}) : \text{card } W(A) = 1\}.$$

Since $\{A \in \mathcal{B}(\mathcal{H}) : \text{card } W(A) = 1\} = \mathbb{C}I$ is nowhere dense in $\mathcal{B}(\mathcal{H})$, it suffices to show $\{A \in \mathcal{B}(\mathcal{H}) : \text{card } W_0(A) > 1\}$ is nowhere dense in $\mathcal{B}(\mathcal{H})$.

We set

$$\mathcal{E} = \{T \in \mathcal{B}(\mathcal{H}) : \|T\| \text{ is an isolated point of } \sigma(|T|) \text{ with dim } \ker(|T| - \|T\|) = 1\}.$$

By Remark 2.1, each $A \in \mathcal{E}$ satisfies card $W_0(A) = 1$. Then it remains to show $\mathcal{E}$ is an open dense subset of $\mathcal{B}(\mathcal{H})$.

The rest of the proof is divided into two claims.

Claim 1. $\mathcal{E}$ is an open subset of $\mathcal{B}(\mathcal{H})$.

It is clear that

$$\mathcal{E} = \{X \in \mathcal{B}(\mathcal{H}) : \|X\| \in \sigma_0(|X|), \dim \mathcal{H}(\|X\|; |X|) = 1\}.$$

Fix an operator $A \in \mathcal{E}$. Clearly we may assume that $A \neq 0$. Choose a Cauchy domain Ω satisfying $\|A\| \in \Omega$ and $\sigma(|A|) \cap \overline{\Omega} = \{\|A\|\}$. Set

$$\alpha = \min\left\{\left\|(\lambda - |A|)^{-1}\right\|^{-1} : \lambda \in \partial\Omega\right\}.$$

Note that the map $X \mapsto |X|$ is continuous. So there exists $\beta > 0$ such that $\||A| - |X|\| < \alpha$ for all $X \in \mathcal{B}(\mathcal{H})$ with $\|A - X\| < \beta$. Furthermore, by Lemma 3.7, we have $\sigma(|X|) \cap \Omega \neq \emptyset$ and $\dim \mathcal{H}(\sigma(|X|) \cap \Omega; |X|) = \dim \mathcal{H}(\|A\|; |A|) = 1$. Then $\mathrm{N}(A, \beta) \subset \mathcal{E}$. This proves Claim 1.

Claim 2. $\mathcal{E}^- = \mathcal{B}(\mathcal{H})$.

We only deal with the case that $\dim \mathcal{H} = \infty$. The proof for the finite dimensional case is similar.

It is clear that

$$|A| = \begin{bmatrix} E & 0 \\ 0 & 0 \end{bmatrix}$$

relative to the decomposition $\mathcal{H} = (\ker A)^{\perp} \oplus \ker A$, where E is injective and positive. Without loss of generality, we assume that $\dim(\ker A)^{\perp} = \infty$.

By [5, Theorem 2.4.4], we can find an injective diagonal operator

$$D = \operatorname{diag}\{a_1, a_2, a_3, \cdots\}$$

relative to some orthonormal basis $\{e_i\}_{i=1}^{\infty}$ of $(\ker A)^{\perp}$ such that $E \cong_a D$ and $a_1 = \|D\|$. Then, given any $\varepsilon > 0$, there exists a unitary $U \in \mathcal{B}((\ker A)^{\perp})$ and $K_0 \in \mathcal{B}((\ker A)^{\perp})$ with $\|K_0\| < 2\varepsilon$ such that

$$U(E + K_0)U^* = \operatorname{diag}\{a_1 + \varepsilon, a_2, a_3, \cdots\} := \widetilde{D}.$$

Assume $A = V|A|$ is the polar decomposition of A. Set

$$B = V \begin{bmatrix} U^*\widetilde{D}U & 0 \\ 0 & 0 \end{bmatrix}. \tag{3.1}$$

Then

$$\|B - A\| = \|U^*\widetilde{D}U - E\| = \|K_0\| < 2\varepsilon.$$

Since $U^*\widetilde{D}U$ is positive and $(\operatorname{ran} U^*\widetilde{D}U)^- = \overline{\operatorname{ran} E}$, it follows that (3.1) is the polar decomposition of B. Note that $\|B\| = a_1 + \varepsilon$ is an isolated point of $|B|$ with $\ker(|B| - a_1 - \varepsilon) = \vee\{e_1\}$. Then $B \in \mathcal{E}$. Since ε was arbitrary, this proves Claim 2. Hence we complete the proof. □

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